Contribution Journal – Module 6: Regression and Classification Models

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Introduction

For this lab, my contribution focused on building and understanding machine learning models using the Titanic dataset. I worked through the steps of preparing the data, training the models, making predictions, and evaluating their performance. My goal was to understand how regression and classification are used to solve different problems.

My Specific Contributions

I worked with the Titanic dataset and used Python to prepare information for the models. For Linear Regression, I used passenger age and class to predict ticket fares. I trained the model, made predictions, and used the Mean Squared Error to evaluate the results. The model had an MSE of 3364.92 and a root mean squared error of about $58.01.

I also worked on the Logistic Regression model to predict whether passengers survived. I encoded the sex column into numerical values and used age, passenger class, and sex as the features. I trained the model and evaluated its predictions using accuracy. The model achieved an accuracy of 74.83%. I also answered the knowledge check questions to explain the differences between regression and classification and why each model uses a different evaluation metric.

My Learning and Reflection

Through my work, I learned that regression predicts numerical values while classification predicts categories. I also learned that selecting the correct evaluation metric is important because MSE measures prediction errors for numerical values, while accuracy measures how many classification predictions are correct. One challenge was understanding what the results meant and how to interpret them. Working through the code helped me connect what I learned in class to a practical example.

Conclusion

Overall, my contribution to this lab involved preparing the Titanic data, training and evaluating Linear Regression and Logistic Regression models, and explaining the results in my own words.

This experience improved my understanding of machine learning and showed me the importance of data preparation, model evaluation, and interpreting results.